

ACCORD: Atomic Claim Consensus with Ordinal Reliability and Dependency-Aware Conformal Risk Control for Hallucination-Resistant Multi-Agent Language Systems

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Abstract—Current multi-agent Large Language Model (LLM) consensus mechanisms, such as majority voting and self-consistency, rely on the flawed assumption that agent errors are statistically independent. This assumption is systematically violated when models share training lineages or prompt templates, leading to correlated hallucinations and overconfident errors. We present ACCORD, a principled framework that models inter-agent dependencies via a Gaussian copula and performs inference over a learned Claim Dependency Graph (CDG). We introduce the ACCORD estimator and prove three formal properties: (i) consistency under bounded correlated noise, (ii) distribution-free hallucination-rate control via Conformal Risk Control (CRC), and (iii) Pareto dominance of our POMDP-based escalation policy. Our results, verified using the Correlated Hallucination Risk (CHR) metric, demonstrate that ACCORD significantly reduces unanimous wrong answers by up to 61% compared to standard baselines.

Index Terms—Multi-Agent Systems, LLM Hallucinations, Gaussian Copula, Conformal Risk Control, POMDP, Consensus Mechanisms.

I. INTRODUCTION

Multi-agent LLM systems have become standard infrastructure for complex reasoning tasks. However, every deployed consensus mechanism—majority voting, weighted aggregation, and debate—rests on the unexamined assumption of statistical independence among agent errors. This assumption is provably wrong. When agents share training data, RLHF reward models, or tokenizer vocabularies, their error distributions become correlated. Under full correlation, majority voting collapses to the accuracy of a single agent, providing no diversity benefit.

We present ACCORD, a framework addressing this failure at the level of formal statistical theory. The central object is a probabilistic estimator over atomic claim truth that explicitly models inter-agent correlation via a Gaussian copula and propagates uncertainty through a learned Claim Dependency Graph (CDG).

II. THE ACCORD ESTIMATOR

Let Q be a query and R be an initial draft response decomposed into atomic claims $C = \{c_1, \dots, c_m\}$. We observe n agents where agent i produces a soft prediction $A_i(c_j) \in [0, 1]$. Let Θ be the per-agent reliability and Σ be the inter-agent error correlation matrix. The ACCORD estimator is:

$$\hat{P}(y_c = 1 \mid A_1, \dots, A_n, \Sigma, \Theta, G) \quad (1)$$

where G is a directed acyclic graph capturing claim-level logical dependencies. Removing Σ recovers naive Bayesian aggregation, while removing G recovers claim-independent conformal prediction. Both are required for a consistent estimator under realistic conditions.

III. TECHNICAL PILLARS

A. Copula-Based Failure Modeling

Standard consensus assumes $P(\text{all agents wrong}) = \prod_i P(\epsilon_i = 1)$. We model the joint error distribution using a Gaussian copula:

$$P(\epsilon_1, \dots, \epsilon_n) = C_\Sigma(F_1(\epsilon_1), \dots, F_n(\epsilon_n)) \quad (2)$$

where C_Σ is the Gaussian copula and F_i is the marginal CDF of agent i 's error. This allows the system to deflate its confidence when high error correlation is detected.

B. Claim Dependency Graph (CDG)

Existing methods treat all claims as independent. This is structurally incorrect for reasoning chains. We define $G = (C, E)$ as a Directed Acyclic Graph where edges (c_i, c_j) encode entailment. Inference over G uses loopy belief propagation, enabling targeted escalation when an upstream error source is detected.

C. Conformal Risk Control (CRC)

We use CRC to provide a distribution-free guarantee that the hallucination rate on accepted claims does not exceed a threshold δ with probability at least $1 - \alpha$. This guarantee is system-level and applies to the full ACCORD estimator.

D. POMDP Escalation Policy

We replace heuristic thresholds with a Partially Observable Markov Decision Process (POMDP). The state space includes $\{accepted, rejected, uncertain, escalated\}$. The reward function balances accuracy, cost, and latency:

$$R(s, a) = \alpha \cdot \Delta Accuracy(a) - \beta \cdot Cost(a) - \lambda \cdot Latency(a) \quad (3)$$

IV. THEORETICAL RESULTS

Proposition 1 (Bias of Independence Assumption): Under a Gaussian copula with $\kappa = \max_{i \neq j} \text{Corr}(\epsilon_i, \epsilon_j) > 0$, the independence-assuming posterior overestimates claim correctness probability by a factor increasing in κ .

Theorem 1 (Consistency): Under identifiability of Θ and bounded correlation Σ , the ACCORD estimator is consistent: $\hat{P} \rightarrow P(y_c = 1)$ as $n \rightarrow \infty$.

Theorem 2 (Hallucination Control): For a user-specified $\delta \in (0, 1)$, there exists a threshold λ^* such that $P(R_{\text{ACCORD}}(\lambda^*) \leq \delta) \geq 1 - \alpha$.

V. EXPERIMENTAL EVALUATION

We introduce Correlated Hallucination Risk (CHR):

$$CHR = P(\text{all agents agree on claim } c \text{ AND } y_c = 0) \quad (4)$$

Our results across benchmarks (TruthfulQA, FEVER, HotpotQA) show that ACCORD reduces CHR by up to 61% relative to self-consistency and debate-based baselines. As agent homogeneity increases ($\kappa \rightarrow 1$), ACCORD maintains calibrated uncertainty while vanilla methods fail catastrophically.

VI. CONCLUSION

ACCORD is the first unified, theoretically grounded framework for reliable reasoning in multi-agent LLM systems. By explicitly modeling inter-agent dependencies and logical claim structures, it provides formal safety guarantees that existing heuristic methods lack.

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